

VU SOLUTION BY RAJA MUSHTAQ:

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CS504 final paper 12-02-2024 by r_m

Solution of. Cs504 sample

Question No: 1

- ***Question:*** Scientific investigations and engineering problem solving require ____ and ____

- ***Options:***

- Calculations, data analysis

- Calculation, knowledge

- Knowledge, solution

- Problem, solution

- ***Correct Answer:*** Calculation, knowledge

Question No: 2

- ***Question:*** Which of the following branch of engineering is based on computer science?

- ***Options:***

- Civil engineering

- Mechanical engineering

- Electrical engineering

- Software engineering

- ***Correct Answer:*** Software engineering

Question No: 3

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- ***Question:*** Software development activities could be performed in a cyclic and that cycle is called_____.

- ***Options:***

- Software activities loop
- Software management cycle
- Software development loop
- Software construction cycle

- ***Correct Answer:*** Software development loop

Question No: 4

- ***Question:*** Which of the following phase is not related to software development's basic phases?

- ***Options:***

- Development
- Vision
- Reconstruction
- Maintenance

- ***Correct Answer:*** Vision

Question No: 5

- ***Question:*** In which of below given phase the software engineer starts designing and coding of the software?

- ***Options:***

- Feasibility study

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- Construction phase
- Development phase
- Engineering phase
- ***Correct Answer:* Development phase**

Question No: 6

- ***Question:*** Insufficient user involvement in requirement gathering leads to ____.

- ***Options:***
 - Quality product
 - Unacceptable products
 - Unacceptable requirement
 - Black hole in system
- ***Correct Answer:* Unacceptable products**

Question No: 7

- ***Question:*** Which of the following techniques, software engineers use in order to understand business domain.

- ***Options:***
 - System models
 - Process model
 - Iterative model
 - Symbolic model
- ***Correct Answer:* Process model**

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Question No: 8

- ***Question:*** Which of the following symbol is used to represent the external entities in DFD?

- ***Options:***

- Squares

- Circle

- Triangle

- Arrow

- ***Correct Answer:*** Circle

Question No: 9

- ***Question:*** In level one of data flow diagram the analyst adds ____ that deal with the external agents.

- ***Options:***

- Data flow symbol

- Internal agents

- Actors

- Processes

- ***Correct Answer:*** Actors

Question No: 10

- ***Question:*** System users often judge a system by its ____ rather than its ____.

- ***Options:***

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- Interface, functionality
- Functionality, Interface
- Functionality, Code
- Interface, Design
- ***Correct Answer:*** Interface, functionality

Question No: 11

- ***Question:*** What does software architecture define?
- ***Options:***
 - The low level structure of the software
 - The detailed structure of the software
 - The high level structure of the software
 - The design of the software
- ***Correct Answer:*** The high level structure of the software

Question No: 12

- ***Question:*** What are the three categories of elements in software architecture?
- ***Options:***
 - Data elements, processing elements, and controlling elements
 - Input elements, processing elements, and output elements

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- Data elements, processing elements, and connecting elements
- Storage elements, processing elements, and linking elements
- ***Correct Answer:*** Data elements, processing elements, and connecting elements

Question No: 13

- ***Question:*** What is Krutchen's architectural view model known as?
- ***Options:***
 - Krutchen's 2+1 architectural view model
 - Krutchen's 3+1 architectural view model
 - Krutchen's 4+1 architectural view model
 - Krutchen's 5+1 architectural view model
- ***Correct Answer:*** Krutchen's 4+1 architectural view model

Question No: 14

- ***Question:*** Which of the following statements is true about the Model View Controller (MVC) pattern and the observer pattern?
- ***Options:***
 - The MVC pattern was based on the observer pattern.

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- The observer pattern was not based on the MVC pattern.
- The MVC pattern and the observer pattern are unrelated.
- The observer pattern was based on the MVC pattern.
- ***Correct Answer:*** The observer pattern was based on the MVC pattern.

Question No: 15

- ***Question:*** Which type of words should never be abbreviated?
- ***Options:***
 - Proper nouns like names of people, places, organizations, and brands
 - Common words listed in a language dictionary
 - Abbreviated forms of words
 - Words used in computing and technology
- ***Correct Answer:*** Proper nouns like names of people, places, organizations, and brands

Question No: 16

- ***Question:*** Which of the following is the correct extension for C++ header files?
- ***Options:***

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- .cpp
- .hpp
- .c
- .h
- ***Correct Answer:* .hpp**

Question No: 17

- ***Question:* What is the recommended extension for C++ source files?**

- ***Options:***

- .c++
- .C
- .cc
- .cpp
- ***Correct Answer:* .cpp**

Question No: 18

- ***Question:* What is the recommended approach to using "magic numbers" in C++ code?**

- ***Options:***

- Keep them as unnamed, hard-coded values
- Replace them with named constants
- Replace them with variables
- Replace them with named variables

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- ***Correct Answer:*** Replace them with named constants

Question No: 19

- ***Question:*** When writing floating point constants in C++, which of the following is recommended?

- ***Options:***

- Without a decimal point
- With a decimal point but no decimal
- With a decimal point and at least one decimal
- With a decimal point and multiple decimals

- ***Correct Answer:*** With a decimal point and at least one decimal

Question No: 20

- ***Question:*** What is the purpose of the following piece of code?

```
a = a >> 2;
```

- ***Options:***

- To shift the value of 'a' two times to the right
- To divide the value of 'a' by 4
- To shift the value of 'a' the times to the left
- To divide the value of 'a' by 2

- ***Correct Answer:*** To shift the value of 'a' two times to the right

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Question No: 21

- ***Question:*** The thin-client model was used to migrate the legacy of _____.

- ***Options:***

- Data-centered Model
- Layered Model
- Reference Model
- Client-server Model

- ***Correct Answer:*** Client-server Model

Question No: 22

- ***Question:*** In ____ architecture contains more than three layers that can run on the processor.

- ***Options:***

- Three-tier Architecture
- N-tier architecture
- Layered Architecture
- Reference Architecture

- ***Correct Answer:*** N-tier architecture

Question No: 23

- ***Question:*** OSI model is an example of _____.

- ***Options:***

- Data Flow Architecture

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- Layered Architecture
- Partitioning the Architecture
- Reference Architecture
- ***Correct Answer:* Layered Architecture**

Question No: 24

- ***Question:* Which one is not the participant of the Observer Pattern?**

- ***Options:***
 - Subject
 - ConcreteSubject
 - ConcreteObserver
 - Object
- ***Correct Answer:* Object**

Question No: 25

- ***Question:* The correct naming convention of C++ is:**

- ***Options:***
 - Int Myage = 25;
 - int myAge = 25;
 - INT MyAge = 25;
 - INT MYAGE = 25;
- ***Correct Answer:* int myAge = 25;**

Question No: 26

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- ***Question:*** The _____ must be declared in individual header files along with the file name matching the class name.

- ***Options:***

- Variables
- Special characters
- Functions
- Classes

- ***Correct Answer:*** Classes

Question No: 27

- ***Question:*** _____ statement contains break and default keywords which are optional.

- ***Options:***

- Switch
- Boolean
- Loops
- Pointers

- ***Correct Answer:*** Switch

Question No: 28

- ***Question:*** The correct example of for loop is _____.

- ***Options:***

- For (int i = 1; i <= 2; i)
- For (int i = 1; i <= 2 < I++)

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- for (int i = 1; i <= 2; i++)
- for (int i = 1; i <= 2; i++)
- ***Correct Answer:** for (int i = 1; i <= 2; i++)

Question No: 29

- ***Question:** Which one is not a special character?
- ***Options:**
 - Tab
 - Back space
 - Page break
 - String
- ***Correct Answer:** String

Question No: 30

- ***Question:** _____ can hold more than a single bit.
- ***Options:**
 - Bit field
 - String
 - Array
 - Pointer
- ***Correct Answer:** Bit field

Question No: 31

- ***Question:** Electric subsystem of house is an example of _____.

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- ***Options:***

- High coupling
- High cohesion
- Low cohesion
- Low energy

- ***Correct Answer:*** High cohesion

Question No: 32

- ***Question:*** In case of _____ everything seems to be connected with everything else.

- ***Options:***

- High coupling
- High cohesion
- Low cohesion
- Low energy

- ***Correct Answer:*** High coupling

Question No: 33

- ***Question:*** Abstraction is the understanding of _____

- ***Options:***

- Internal logic
- Internal interface
- External interface
- External environment

- ***Correct Answer:*** External interface

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Question No: 34

- ***Question:*** The properties of an object are usually

- ***Options:***

- Attractive

- Static

- Dynamic

- Controlled

- ***Correct Answer:*** Dynamic

Question No: 35

- ***Question:*** Objects do not exist in _____

- ***Options:***

- Models

- Parts

- Isolation

- Encapsulation

- ***Correct Answer:*** Isolation

Question No: 36

- ***Question:*** The intent of OOA is to define all classes, their relationships, and their _____

- ***Options:***

- Scope

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- Behavior
- Model
- Instance
- ***Correct Answer:* Behavior**

Question No: 37

- ***Question:* OOD transforms the analysis model into**

-
- ***Options:***
 - Unit model
 - Behavioral model
 - Design model
 - Testing model
 - ***Correct Answer:* Design model**

Question No: 38

- ***Question:* _____ is a notation used for documenting the object-oriented design.**

- ***Options:***
 - UMS
 - Rumbaugh
 - Board
 - C++
- ***Correct Answer:* Rumbaugh**

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Question No: 39

- ***Question:*** _____ is not a notation used for documenting the object-oriented design.

- ***Options:***

- UML
- Rumbaugh
- Coad
- C++

- ***Correct Answer:*** C++

Question No: 40

- ***Question:*** Once all the model components have been identified, we will eliminate the _____ components.

- ***Options:***

- Redundant components
- Relevant components
- Board components
- Major components

- ***Correct Answer:*** Redundant components

Question No: 41

- ***Question:*** Identify below given statement as correct or incorrect, justify your answer in either case. "User requirements are written from a system perspective".

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- *Solution:* This statement is incorrect. User requirements should be written from the perspective of the users who will be using the system, not from the perspective of the system itself. User requirements focus on what the users need the system to do, their preferences, and their expectations, rather than technical details about the system.

Question No: 42

- *Question:* You are required to identify use cases and actors from the below given use case diagram.
- *Solution:* (Solution depends on the provided use case diagram)

Question No: 43

- *Question:* In Data-Centered (Repository) model, sub-systems need to exchange information and data. Write the two ways in which this exchange of the data can be handled?
- *Solution:* In the Data-Centered model, data exchange between subsystems can be handled through shared databases or through message passing mechanisms such as APIs or web services.

Question No: 44

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- ***Question:*** What is the appropriate use of the term "initialize"?
- ***Solution:*** "Initialize" is used in programming to set the initial value of a variable or data structure before it is used in the program. It ensures that the variable or data structure starts with a known state before any operations are performed on it.

Question No: 45

- ***Question:*** Statement: `int myBooks[5] = {1, 2, 3, 4, 5}`
Identify the coding mistake and correct it.
- ***Solution:*** The coding mistake is that the array initialization is incorrect. Since arrays are zero-indexed in C++, the correct initialization should start from index 0. So it should be: `int myBooks[5] = {0, 1, 2, 3, 4};`

Question No: 46

- ***Question:*** A program contains few logical errors. How it is resolved by a programmer?
- ***Solution:*** Logical errors are resolved by carefully reviewing the program's logic, identifying the incorrect or unintended behavior, and then modifying the code to correct the logic. This often involves using debugging techniques such as printing intermediate values,

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stepping through the code with a debugger, or writing test cases to isolate the issue.

Question No: 47

- *Question:*** Enlist at least five components of the Code View, which is basically an extension of the development view.
- *Solution:*** (Solution depends on the specific components defined in the Code View)

Question No: 48

- *Question:*** Two main documents produced during requirement engineering phase, are Requirement Statement and Requirement Specification. You are required to write down any five characteristics of good requirements statement document.
- *Solution:***
 - 1. Clear and unambiguous language**
 - 2. Measurable and testable criteria**
 - 3. Complete coverage of user needs**
 - 4. Consistency with stakeholder expectations**
 - 5. Traceability to higher-level requirements and project goals**

Question No: 49

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- ***Question:*** A software house wants to launch an E-commerce website, but they want to perform software testing. They choose unit testing for the website. Is unit testing helpful in a small-scale organization elaborate with a reason? Enlist any two benefits.

- ***Solution:*** Yes, unit testing can be helpful in a small-scale organization for several reasons:

1. Early bug detection: Unit tests can catch bugs at an early stage of development, allowing them to be fixed before they escalate into larger issues.

2. Improved code quality: Unit tests encourage developers to write modular and well-structured code, leading to higher-quality software.

Question No: 50

- ***Question:*** A Student class has following attributes and operations. Convert the "Student" class in the form of diagram using UML notation for classes. Attributes a. Roll no. b. Name c. Subject Operations a. AttemptQuiz() b. SubmitAssignment() c. PayFee()

- ****Solution:**

Here's the UML class diagram for the "Student" class: _____

```
|      Student      |  
|-----|
```

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```
| - rollNo : int      |  
| - name : string    |  
| - subject : string  |  
|-----|  
| + attemptQuiz()    |  
| + submitAssignment() |  
| + payFee()         |
```

-----In the diagram:Attributes are represented as private variables (-) with their data types.Operations are represented as public methods (+) with their names and parameters (if any). جزاک اللہ خیرا.

کثیرا واحسن الجزاء فی الدنیا والآخرۃ

Thynks... 🍀 ❤️

Regards:r_m... 🤖